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Peer-to-Peer Carpooling & Expense Splitter

Requirements Specification and Use-Case Documentation

1. Project Overview

A commuter matching portal that matches drivers and passengers with overlapping daily routes, enforces detour tolerance thresholds, and calculates an automated fuel cost share split.

Target Stakeholders / Actors: Rider/Passenger, Driver Host

2. Functional Requirements (FR)

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall match drivers and passengers with overlapping daily routes.	High	A match is shown when routes overlap and the detour is within 1.5 km.	Helps commuters find suitable carpool partners.
FR-002	Functional	The system shall allow users to create and manage their commute profiles.	High	Users can add, edit, and save pickup location, destination, and travel time.	Required to find suitable carpool matches.
FR-003	Functional	The system shall allow drivers to create carpool rides and specify available seats.	High	Drivers can publish a ride with route, time, and number of available seats.	Allows drivers to offer rides to passengers.
FR-004	Functional	The system shall allow passengers to request or	High	A passenger can send a request and the driver	Enables ride sharing between users.

		join an available ride.		can accept or reject it.	
FR-005	Functional	The system shall calculate and split the fuel cost among ride participants.	Medium	The total fuel cost and each participant's share are displayed correctly.	Makes expense sharing fair and easy.

3. Non-Functional Requirements (NFR)

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The system shall verify user identity documents and store travel history securely with encrypted backups.	High	Identity verification works correctly and sensitive data is securely stored.	Protects users and their personal information.
NFR-002	Usability	The system shall provide a simple and easy-to-use interface for drivers and passengers.	Medium	Users can perform major tasks without difficulty.	Improves user experience and adoption.

4. UML Use-Case Diagram Specification

Actors:

- Passenger / Rider
- Driver Host

Primary Use Cases:

- Register / Login
- Manage Commute Profile
- Find Carpool Match
- Create Ride
- Request to Join Ride
- Accept / Reject Request
- Calculate Expense
- Split Fuel Cost

Required Relationships:

- <<include>>: Request to Join Ride includes Find Carpool Match.
- <<extend>>: Split Fuel Cost extends Calculate Expense.

5. Use-Case Documentation

Use Case: Request to Join Ride

Preconditions

- The user is registered and logged into the system.
- A suitable carpool ride is available.

Postconditions

- The ride request is sent to the driver.
- The passenger is notified about the request status.

Main Success Scenario

1. Passenger logs into the system.
2. Passenger searches for available carpool rides.
3. The system displays matching rides.
4. Passenger selects a suitable ride.
5. Passenger sends a request to join.
6. The system sends the request to the driver.
7. The driver accepts the request.
8. The system confirms the passenger's seat.

Alternate Flow

- If no suitable ride is available, the system displays a 'No matching rides found' message.
- If the driver rejects the request, the system notifies the passenger.
- If all seats are full, the passenger cannot join the ride.

UML DIAGRAM

